FRANCO SILVESTRI

+1 (786) 823-5805 | fransilvestri02@gmail.com | colo2121.github.io | linkedin.com/in/franco-silvestri-98022b212

EDUCATION

Florida International University — Miami, FL, USA

BS in Computer Science, Minor in Mathematical Sciences | GPA: 3.95/4.0 | Honors College — Aug 2023 - May 2027

Accelerated BS/MS (4+1): returning to FIU for the MS in Computer Science, enrolled through 2028 — Aug 2027 - May 2028

Relevant coursework: Data Structures & Algorithms, Advanced Databases, Machine Learning, Artificial Intelligence

First-author poster, FIU Undergraduate Research Conference 2026: real-time mercury characterization with mobile robots

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript/TypeScript, SQL

Systems and Backend: Distributed Systems, Real-Time Systems, Networking, ROS 2, DDS Middleware, Django REST, PostgreSQL

Infrastructure and DevOps: Docker, Kubernetes, GitLab CI, Jenkins, Linux, AWS

ML and Perception: PyTorch, GPyTorch, CUDA, TensorRT, Jetson, Semantic Segmentation, SLAM, Nav2, Sensor Fusion, Gaussian Processes

EXPERIENCE

Robotics Research Assistant, Perception and Systems Lead — Jan 2024 - Present

FIU Applied Research Center | DOE MSIPP project with Savannah River National Lab and Oak Ridge Y-12 | Miami, FL, USA

- Designed a sensor-agnostic, modular ROS 2 node architecture so new sensors integrate without touching downstream navigation, mapping, or visualization, enabling deployment across multiple robotic platforms
- Cut real-time inference latency 22x over a PyTorch GPU baseline (CUDA, quantization, TensorRT) for a semantic-segmentation pipeline on a Boston Dynamics SPOT, sharing one Jetson Orin Nano GPU with a Gaussian Process; fine-tuned DeepLabV3+/STEGO to a maximum of 72% mIoU across 21 navigation-critical classes
- Built distributed ROS 2 components in C++/Python and designed the robot's networking from scratch (IP addressing, low-level Linux, DDS tuning, traffic prioritization), sustaining concurrent lidar, SLAM, inference, and AR streams across onboard and offboard compute
- Designed a multi-sensor fusion pipeline in C++ merging Velodyne lidar with six onboard depth cameras into a unified point cloud through calibrated transforms; implemented SLAM with Nav2 planning and open-frontier exploration of unknown environments
- Built an online Gaussian Process regression node (GPyTorch, CUDA) exposing incremental feed and predict ROS 2 services, fine-tuned to the live CO_2 sensor response and fused with SLAM odometry into real-time contaminant maps with uncertainty estimates

Genworth Financial, IT Development Program — Summers 2025 and 2026

Richmond, VA, USA | Returning intern, second consecutive term, rotated to the Architecture team

Platform Engineering and DevOps, Architecture Team — May 2026 - Present

- Standardized CI/CD across a 100+ system enterprise portfolio, building end-to-end pipelines for 80+ via reusable GitLab CI templates per app type (React/Angular front ends, Node/Yarn builds, Java REST APIs, monoliths), rebuilding from scratch where legacy infrastructure was broken
- Raised the average internal DevOps Adoption score across a 110+ project portfolio from 39 to 75, top-ranked among teams maintaining the largest system portfolios
- Integrated automated security gates into every pipeline (SAST, DAST, dependency scanning), partnering with the Middleware team to map each application's authentication structure, including permissions, tokens, and protected routes, so dynamic scans could execute against secured endpoints
- Automated artifact promotion and deployment to servers with Jenkins; containerized workloads with Docker and tuned Kubernetes pod resource allocation per system, partnering with each application's developers and architecture leadership on the right approach per system

Application Development, IT Sourcing and Supply Management — May 2025 - Aug 2025

- Owned an enterprise supplier-feedback product through the full development cycle on Microsoft Power Platform: requirements, data architecture, build, enterprise security and compliance review, release, and knowledge transfer
- Replaced manual Outlook tracking and 100+ per-supplier Excel workbooks for a 50+ person department; centralized analytics and automated reminders cut feedback collection from hours to minutes

PROJECTS

Pharmacy Quoting and Compliance SaaS, Co-founder and Sole Backend Engineer — Feb 2023 - Present

- Sole backend engineer on a 2-person team, owning architecture, implementation, testing, and deploy readiness (Python, Django REST, PostgreSQL, Docker, CI/CD with automated migrations and health-gated rollouts); 11 domain modules with a database-free core for exact-Decimal financial math and immutable, version-stamped quote snapshots; preparing pilot launch
- Designed a 5-layer authorization pipeline: tenant isolation (cross-tenant access returns 404, not 403, to block resource enumeration), plan entitlement gating, capability-based RBAC with default-deny and instant revocation, state-machine transition guards, and field-level redaction of sensitive pricing, all enforced through one central hook so no endpoint can bypass it
- Architected an append-only audit event store as the single source of truth for all state changes; reporting and compliance features are read-only projections over the log and never touch domain tables
- Built a 2,900-test suite across unit, integration, and end-to-end levels, including a generated role by endpoint by method RBAC matrix that fails CI when any endpoint ships without a classified capability, preventing authorization drift

LEADERSHIP

INIT FIU, Robotics Technical Lead, E-Board — Jan 2025 - Present

Led 16+ embedded systems and robotics workshops (video playlist); co-organized ShellHacks, Florida's largest hackathon

Panther Robotics, Software Team Lead — Aug 2023 - Dec 2024

Led a 7-member team exploring reinforcement learning for robotic arm control in simulation (Keras, PyBullet, Isaac Sim)